Chapter 3.

Finance I: Saving and borrowing.

Revision of simple interest, compound interest and recursive rules.

The Preliminary Work at the beginning of this book reminded you of simple interest, compound interest and sequences defined by giving the first term and a recursive rule. The next examples and exercise 3A that follows should serve as a further reminder.

Note: The *Simple Interest*, \$I, earned when \$P is invested for T years in an account paying R% per annum simple interest is given by $I = \dfrac{PRT}{100}$

If instead we use R in decimal form we use $I = PRT$.

For example if the interest rate is 6% the first formula would use R = 6 but the second would use R = 0·06.

Example 1

How much interest would be earned in 4 years on an investment of \$5000 at 6% per annum simple interest.

Using $I = PRT$ with P = 5000, R = 0·06 and T = 4

$$\begin{aligned} I &= 5000 \times 0{\cdot}06 \times 4 \\ &= 1200 \end{aligned}$$

The interest earned will be \$1200.

Example 2

What annual rate of simple interest will see an investment of \$8000 earn \$3600 simple interest in 6 years?

Using $I = PRT$ with I = 3600, P = 8000 and T = 6

$$\begin{aligned} 3600 &= 8000 \times R \times 6 \\ 3600 &= 48000R \\ R &= \frac{3600}{48000} \\ &= 0{\cdot}075 \end{aligned}$$

The required rate of simple interest is 7·5%.

Example 3

A simple interest arrangement of 5·4% per annum sees an initial investment grow to \$11121·60 in 6 years. What was the initial investment?

Using $I = PRT$ with $P + I = 11121{\cdot}60$, $R = 0{\cdot}054$ and $T = 6$

$$\begin{aligned} P + I &= P + PRT \\ 11121{\cdot}60 &= P + P \times 0{\cdot}054 \times 6 \\ 11121{\cdot}60 &= P + 0{\cdot}324P \\ 11121{\cdot}60 &= 1{\cdot}324P \\ P &= 8400 \end{aligned}$$

The initial investment was \$8400.

Example 4

\$25000 is invested for 4 years with an annual compound interest rate of 6%. Find the amount this account is worth at the end of the 4 years if the compounding occurs

(a) annually,
(b) quarterly,
(c) daily.

(a) Compounding annually.
There will be 4 compoundings in the 4 years, each of 6%.

$$\begin{aligned} \text{Value after 4 years} &= \$25000 \times 1{\cdot}06 \times 1{\cdot}06 \times 1{\cdot}06 \times 1{\cdot}06 \\ &= \$25000 \times 1{\cdot}06^4 \\ &= \$31561{\cdot}92 \text{ to the nearest cent.} \end{aligned}$$

(b) Compounding quarterly.
There will be 4 compoundings each year, each of 1·5% (= 6% ÷ 4).
I.e. 16 in the 4 years

$$\begin{aligned} \text{Value after 4 years} &= \$25000 \times 1{\cdot}015^{16} \\ &= \$31724{\cdot}64 \text{ to the nearest cent.} \end{aligned}$$

(b) Compounding daily.
There will be 365 compoundings each year, each of (6% ÷ 365).
I.e. 1460 (= 365 × 4) in the 4 years

$$\begin{aligned} \text{Value after 4 years} &= \$25000 \times \left(1 + \frac{0{\cdot}06}{365}\right)^{365 \times 4} \\ &= \$31780{\cdot}60 \text{ to the nearest cent.} \end{aligned}$$

Note:
- For convenience, in this book, we will ignore leap years and assume that all years have 365 days.
- Prior to the ready availability of computers and calculators some banking calculations, again for convenience, were based on a concept called *a banker's year*. This concept takes a year as being 360 days and consisting of twelve equal months each of thirty days. This concept is mentioned here for information only, it will not be used in this course.

Example 5

At an 8% annual compound interest rate, with compounding every six months, how many years would it take for an initial investment of \$2500 to grow to \$3700 ?

There will be 2 compoundings each year, each of 4%.

If the account runs for T years:

$$\$3700 = \$2500 \times 1{\cdot}04^{2T}$$

Using the solve facility of some calculators gives T = 4·998 (rounded to 3 dp)

It would take 5 years.

Consider an investment of \$8000 at 5% per annum simple interest. The annual value of this investment forms the sequence:

\$8000 \$8400 \$8800 \$9200 \$9600 ...

I.e. an arithmetic progression with first term \$8000 and common difference \$400.

Recursive rule: $T_1 = \$8000$, $T_{n+1} = T_n + \$400$.

Or, if we want T_n to represent the value after n years, $T_0 = \$8000$, $T_{n+1} = T_n + \$400$.

Had the investment been at 5% compound interest, with interest compounded annually, the sequence would be:

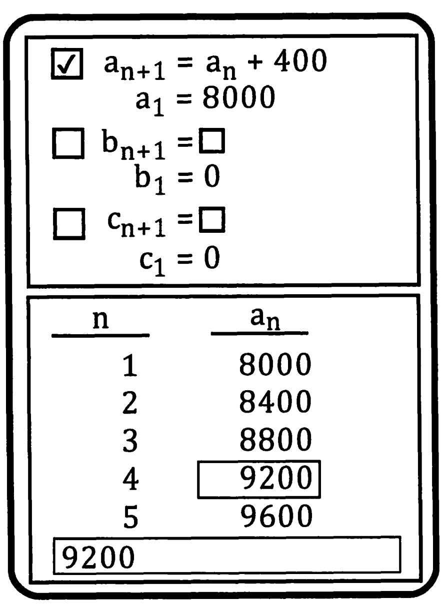
\$8000 \$8400 \$8820 \$9261 \$9724·05 ...

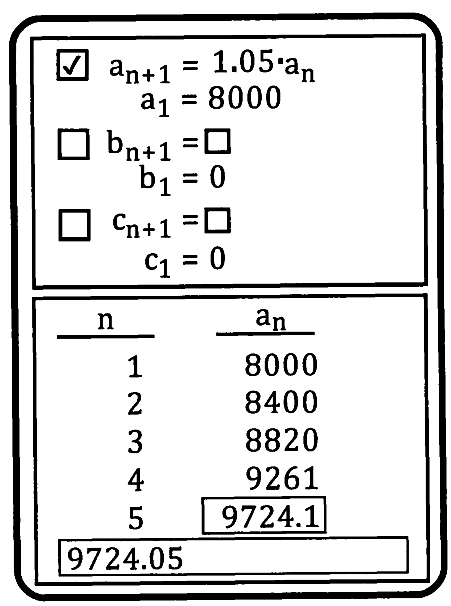
I.e. a geometric progression with first term \$8000 and common ratio 1·05.

Recursive rule: $T_1 = \$8000$, $T_{n+1} = 1{\cdot}05 \times T_n$.

Or, if we want T_n to represent the value after n years, $T_0 = \$8000$, $T_{n+1} = 1{\cdot}05 \times T_n$.

In this way recursive rules can be useful for simple and compound interest situations.





Exercise 3A

Simple interest.

1. What will be the value after 6 years of \$2000 invested at 6% per annum simple interest?

2. How long will it take for \$800, invested in an account paying 7·5% per annum simple interest, to become \$1100?

3. What annual rate of simple interest is needed to see an investment of \$5000 become \$7500 in 4 years?

4. How much must have been invested in an account paying 8% simple interest if after 6 years the total interest earned was \$1152?

5. What annual rate of simple interest is needed to see an initial investment of \$6600 become \$7007 in 10 months.

6. A simple interest arrangement of 6·5% per annum sees an initial investment grow to \$10703 in 6 years. What was the initial investment?

Compound interest.

7. \$4000 is invested at 8% per annum compounded annually. Determine the value of this investment after 6 years.

8. \$500 is left in an account paying 6% per annum compounded six monthly. How much will this account be worth 50 years later?

9. How much interest is earned on \$50000 invested for 4 years if the interest rate is 6% per annum compounded annually?
   How much more interest would be earned in the 4 years if instead the interest is compounded monthly?

10. How much interest is earned on \$400000 invested for ten years if the interest rate is 8% per annum compounded annually?
    How much more interest would be earned in the ten years if instead the interest is compounded daily?

11. How much must be invested now into an account paying 7·5% compound interest, compounded annually, for the account to be worth \$10000 in five years?

12. If an amount is invested at 6% per annum compound interest, with interest compounded quarterly, how long would it take for the initial investment to double in value?

13. At a 9% annual compound interest rate, with compounding every six months, how long would it take for an initial investment of \$5000 to first exceed \$12000?

Recursive rules.

Determine the first five terms and, with the help of a calculator that is able to handle recursive rules if necessary, the fifteenth term, of each of the following recursively defined sequences.

14. $T_{n+1} = T_n + 3, \qquad T_1 = 5.$

15. $T_{n+1} = T_n - 3, \qquad T_1 = 5.$

16. $u_{n+1} = u_n + 5, \qquad u_1 = -10.$

17. $a_n = a_{n-1} + 2{\cdot}5, \qquad a_1 = 12{\cdot}5.$

18. $T_{n+1} = 2 \times T_n, \qquad T_1 = 0{\cdot}25.$

19. $T_{n+1} = 1{\cdot}5 \times T_n, \qquad T_1 = 20\,480.$

20. $a_{n+1} = 0{\cdot}5 \times a_n, \qquad a_1 = 2\,621\,440.$

21. $u_{n+1} = 0{\cdot}5 \times u_n, \qquad u_1 = 2^{30}.$ (Give answers as powers of 2.)

22. $T_{n+1} = 2 \times T_n + 3, \qquad T_1 = 5.$

23. $T_{n+1} = 1{\cdot}2 \times T_n + \$100. \qquad T_1 = \$1000.$ (Write T_{15} to the nearest cent.)

24. $T_{n+1} = 1{\cdot}05 \times T_n - \$250. \qquad T_1 = \$5000.$

25. $T_n = a_n + u_n$ where $a_{n+1} = a_n + 6, \qquad a_1 = 5$
and $u_{n+1} = u_n + 5, \qquad u_1 = 7.$

26. $T_n = a_n \times u_n$ where $a_{n+1} = 2 \times a_n, \qquad a_1 = 5$
and $u_{n+1} = 0{\cdot}5 \times u_n, \qquad u_1 = 163\,840.$

27. $T_{n+2} = T_{n+1} + T_n, \qquad T_1 = 1, \qquad T_2 = 1.$

28. $T_{n+2} = 2 \times T_{n+1} + T_n, \qquad T_1 = 2, \qquad T_2 = 3.$

29. $T_{n+1} = (-1)^n\, T_n, \qquad T_1 = 5.$

30. $T_{n+1} = (-1)^n \times T_n + 2, \qquad T_1 = 5.$

Effective annual interest rate.

Consider \$100 invested in an account paying 6% per annum compound interest.
After one year, with compounding ...

occurring annually, the account will be worth $\$100 \times 1{\cdot}06 = \$106$

occurring six monthly, the account will be worth $\$100 \times 1{\cdot}03^2 = \$106{\cdot}09$

occurring quarterly, the account will be worth $\$100 \times 1{\cdot}015^4 = \$106{\cdot}136$ (correct to 3 dp)

occurring monthly, the account will be worth $\$100 \times 1{\cdot}005^{12} = \$106{\cdot}168$ (correct to 3 dp)

occurring daily, the account will be worth $\$100 \times \left(1 + \frac{0{\cdot}06}{365}\right)^{365} = \$106{\cdot}183$ (correct to 3 dp)

Thus whilst 6% per annum returns \$6 interest in the year when compounded annually, more frequent compoundings sees our 6% interest rate return more than \$6 on \$100 invested, i.e. more than 6%. The 6% is said to be the **nominal annual interest rate** but to make comparison between the effect of the different compounding periods we call the 6·09%, 6·136%, 6·168% and 6·183% returns shown above the **effective annual interest rate**s.

Thus for a nominal compound interest rate of 6% per annum the effective annual interest rate for six monthly compounding is 6·09%. The effective annual interest rate for quarterly compounding is 6·136% (correct to three decimal places), the effective annual interest rate for monthly compounding is 6·168% (correct to three decimal places) etc.

This effective annual interest rate can be determined by consideration of the compound interest earned in one year by an investment of \$100, as shown above, or by use of the following formula:

For a nominal interest rate of i per annum and n compounding periods per year,

$$\text{effective annual interest rate} = \left(1 + \frac{i}{n}\right)^n - 1.$$

This is sometimes written as

$$i_{\text{effective}} = \left(1 + \frac{i}{n}\right)^n - 1.$$

Readers should check that with suitable rounding this formula does indeed give 0·0609 (i.e. 6·09%), 0·06136 (i.e. 6·136%) and 0·06168 (i.e. 6·168%), for $i = 0{\cdot}06$ and values for n of 2, 4 and 12 respectively, as quoted above.

Example 6

Find the effective annual interest rate for a nominal interest of 8% per annum with compounding occurring quarterly.

By considering $100.

8% per annum is 2% per quarter.

After 1 year a $100 investment

becomes $\$100 \times 1{\cdot}02^4$

$= \$108{\cdot}243$ (correct to 3 dp)

By the formula.

Using $i_{\text{effective}} = \left(1 + \frac{i}{n}\right)^n - 1$

with $i = 0{\cdot}08$

and $n = 4$.

$$i_{\text{effective}} = \left(1 + \frac{0{\cdot}08}{4}\right)^4 - 1 = 0{\cdot}08243216.$$

The effective annual interest rate is 8·243% (correct to 3 dp).

Note • **If a question simply refers to an annual interest rate of, say, 8%, it should be assumed that it is the nominal annual rate that is being quoted.**

Exercise 3B

1. Copy and complete the following table. (1 yr = 12 months = 52 weeks = 365 days.)

Compounding frequency.	Nominal annual interest rate (%).	Effective annual interest rate (%).
Annual	4%	
Six monthly	4%	
Quarterly	4%	
Monthly	4%	
Weekly	4%	
Daily	4%	

2. Copy and complete the following table. (1 yr = 12 months = 52 weeks = 365 days.)

Compounding frequency.	Nominal annual interest rate (%).	Effective annual interest rate (%).
Annual	8%	
Six monthly	8%	
Quarterly	8%	
Monthly	8%	
Weekly	8%	
Daily	8%	

3. Looking at your answers for the effective annual interest rates in questions 1 and 2, are the rates for the 8% situation simply twice those of the 4% situation?
If not, why not?

Does your calculator have a nominal interest ↔ effective interest facility?
Are there any "effective interest rate calculators" on the internet?
Investigate.

Another type of interest rate, which at the time of writing is not specifically mentioned in the syllabus for this unit, is the **real interest rate**.
Just as the effective rate is the rate once the effects of compounding have been allowed for, the real interest rate is the rate once inflation has been allowed for.
Investigate.

Initial deposit plus regular investments.
The investments considered so far in this chapter have all involved a "one off" investment deposited in an account and left to earn interest until some later date. In practice many savings schemes involve an initial amount being invested followed by further deposits into the account, perhaps on a regular basis. Consider for example an initial investment of $5000 into an account paying compound interest of 6% per annum, compounded annually, with an extra $500 deposited into the account each year thereafter.

Initial deposit	=	$5000
Amount in account at end of 1 yr	=	$5000 × 1·06 + $500
	=	$5800
Amount in account at end of 2 yrs	=	$5800 × 1·06 + $500
	=	$6648
Amount in account at end of 3 yrs	=	$6648 × 1·06 + $500
	=	$7546·88

With T_1 = $5000, T_2 = $5800, T_3 = $6648, T_4 = $7546·88 etc, what we have here is a sequence that can be defined recursively by the rule:

$$T_{n+1} = 1{\cdot}06 \times T_n + \$500. \qquad T_1 = \$5000.$$

Or, if we want the end of year number to be the same as the term number we use

$$T_{n+1} = 1{\cdot}06 \times T_n + \$500. \qquad T_0 = \$5000.$$

Then T_1 will give the value at the end of year one, T_2 will give the value at the end of year two, T_3 will give the value at the end of year three, etc.

The progress of the investment can then be easily viewed using a calculator with an ability to display the terms of a recursively defined sequence, or using a computer spreadsheet – see the displays on the next page.

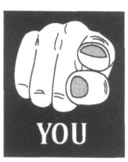
☑ $a_{n+1} = 1.06 \cdot a_n + 500$
$a_1 = 5000$
☐ $b_{n+1} = \square$
$b_1 = 0$
☐ $c_{n+1} = \square$
$c_1 = 0$

n	a_n
4	7546.9
5	8499.7
6	9509.7
7	10580.
8	$\boxed{11715.}$

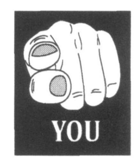
11715.0701198851

	A	B	C	D
1	Initial investment			\$5,000.00
2	Interest rate per yr (%)			6.00
3	Annual deposit			\$500.00
4	Balance at end of year		1	\$5,800.00
5			2	\$6,648.00
6			3	\$7,546.88
7			4	\$8,499.69
8			5	\$9,509.67
9			6	\$10,580.25
10			7	\$11,715.07
11			8	\$12,917.97
12			9	\$14,193.05
13			10	\$15,544.64

- Remember that in the display above left, with the first term being the value after zero years, the 8th term will be the value of the investment after 7 years.



Create a spreadsheet like the one shown above right. Try to make your spreadsheet adaptable to other situations of this type by making it such that by simply changing the amounts in cells D1, D2 and D3 the annual balances are automatically recalculated.



Example 7

At the beginning of one month \$1000 is invested into an account paying interest at 7·5% per annum, compounded monthly, and an extra \$100 is invested at the end of that first month and the end of every month thereafter. How much is the account worth at the end of the month two years later, just after the \$100 deposit for that month is made? (For the purposes of this question assume that a year consists of 12 equal months.)

Initial value = \$1000

Value at end of 1st month = $\$1000 \times \left(1 + \frac{0{\cdot}075}{12}\right) + \$100$ I.e. \$1106·25

Value at end of 2nd month = \$1106·25 × 1·00625 + \$100

Thus with T_n the value of the investment at the end of the nth month

$$T_{n+1} = 1{\cdot}00625 \times T_n + \$100. \qquad T_0 = \$1000.$$

Either using the facility of some calculators to display the terms of recursively defined sequences, or by using your version of the spreadsheet shown above right, with appropriate entries suitably changed (see next page):

Value at end of 24th month = T_{24}

= \$3741·96 (nearest cent).

Hence two years after the initial investment, and just after the \$100 deposit for that month has been made, the account will be worth \$3741·96.

☑ $a_{n+1}=1.00625 \cdot a_n+100$
$a_0=1000$
☐ $b_{n+1}=\square$
$b_0=0$
☐ $c_{n+1}=\square$
$c_0=0$

n	a_n
20	3256.0
21	3376.4
22	3497.5
23	3619.3
24	$\boxed{3742.0}$

3741.9643079925

	A	B	C	D
1	Initial investment			$1,000.00
2	Interest rate per month (%)			0.625
3	Monthly deposit			$100.00
4	Balance at end of month		1	$1,106.25
5			2	$1,213.16
6			3	$1,320.75
7			4	$1,429.00
8			5	$1,537.93
9			6	$1,647.54
10			7	$1,757.84
11			8	$1,868.83
12			9	$1,980.51
13			10	$2,092.89
14			11	$2,205.97
15			12	$2,319.75
16			13	$2,434.25
17			14	$2,549.47
18			15	$2,665.40
19			16	$2,782.06
20			17	$2,899.45
21			18	$3,017.57
22			19	$3,136.43
23			20	$3,256.03
24			21	$3,376.38
25			22	$3,497.48
26			23	$3,619.34
27			24	$3,741.96

Some calculators have built in financial programs that allow some standard financial calculations to be performed automatically, given appropriate information.

The display on the right shows typical information that would be displayed for the situation just encountered.

The N in the display is the number of investment periods, in this case 24 months.

I(%) is the annual interest rate as a percentage, in this case 7·5.

The initial investment of $1000 is shown as PV, for present value. This is shown as a negative amount because from the investor's point of view it is money being deposited (outgoing), not money being received.

The additional investment per month (Pmt) is $100, and again from the investor's point of view this is shown as a negative (outgoing).

Compound Interest

N	24
I(%)	7.5
PV	-1000
Pmt	-100
$\boxed{\text{FV}}$	3741.964308
PpY	12
CpY	12

Payments per year and compoundings per year are each 12.

Provided the program has the correct settings, eg the payment date set as being at the end of each compounding period not at the beginning, when asked to determine the final value (FV) the calculator returns the final value as $3741·96, as before.

Exercise 3C For questions 1 and 2 use recursion. For 3, 4 and 5 do each question <u>twice</u>, once using recursion and once using a calculator with financial capability.

1. \$4000 is invested into an account paying interest at 8% per annum, compounded annually, and an extra \$200 is invested at the end of each 12 months. Thus:
   Amount in account at end of 1 yr $= \$4000 \times 1{\cdot}08 + \$200 \qquad \leftarrow T_1$
   Amount in account at end of 2 yrs $= (\$4000 \times 1{\cdot}08 + \$200) \times 1{\cdot}08 + \$200 \qquad \leftarrow T_2$
   Express T_{n+1} in terms of T_n and determine (nearest cent) the amount in the account at the end of ten years, after the \$200 for that year has been added.

2. \$5000 is used to open an investment account paying interest at 6% per annum, compounded monthly, and an extra \$100 is invested at the end of each one month period thereafter. Thus, the value of the account progresses as follows:
   Initially $= \$5000 \qquad \leftarrow T_0$
   At the end of 1 month $= \$5000 \times 1{\cdot}005 + \$100 \qquad \leftarrow T_1$
   At the end of 2 months $= (\$5000 \times 1{\cdot}005 + \$100) \times 1{\cdot}005 + \$100 \qquad \leftarrow T_2$
   (a) Express T_{n+1} in terms of T_n.
   (b) The account is closed at the end of three years, without the \$100 for the end of the final month being made. Determine the final value of the account.

3. At the beginning of 2014 Tenielle opens a savings account by depositing \$2500 into an account paying 8% per annum, compounded annually. Tenielle plans to follow that initial deposit with further deposits each of \$1000 made at the end of every period of one year thereafter.
   Assuming Tenielle keeps to her plans with regards to the further deposits, how much is the account worth at the beginning of 2025 (i.e. just after the \$1000 deposit for the end of 2024 has been made)?

4. At the beginning of 2014 Sanchez opens a savings account by depositing \$500 into an account paying 7·8% per annum, compounded annually. Sanchez plans to follow that initial deposit with further deposits each of \$500 made at the end of every period of one year thereafter.
   Assuming Sanchez keeps to his plans with regards to the further deposits, how much is the account worth at the end of 2020, i.e. just before the \$500 deposit is made?

5. At the beginning of one month \$5000 is invested into an account paying interest at 9% per annum, compounded monthly, and an extra \$200 is invested at the end of that month and the end of every month thereafter. How much is the account worth at the end of the month three years later, just after the \$200 deposit for that month?

6. \$5000 is used to open an account paying interest that is compounded annually, and an additional \$1000 is to be added to the account at the end of every one year period thereafter. Using a spreadsheet, or a calculator able to display the terms of a recursively defined sequence, or financial programs available on some calculators or internet sites, find the annual interest rate required for this account to be worth (a) \$10 000 after 3 years (including the final \$1000 payment),
   (b) \$100 000 after 25 years (including the final \$1000 payment).

Compounding isn't only about investing.

In a financial context, compounding is not only applied when we invest money, it is also applied to the interest we owe on borrowed money. For example, suppose you borrow \$50 000 from a bank or financial institution. You will be charged interest on this loan. Let us suppose that the interest is charged at the rate of 6% per annum nominal rate, compounded annually.

After one year you will owe	$\$50000 \times 1{\cdot}06$	i.e. \$53 000	
After two years you will owe	$\$50000 \times 1{\cdot}06^2$	i.e. \$56 180	
After three years you will owe	$\$50000 \times 1{\cdot}06^3$	i.e. \$59 550·80	etc.

If instead the compounding occurred monthly:

After one year you will owe	$\$50000 \times 1{\cdot}005^{12}$	i.e. \$53 083·89	
After two years you will owe	$\$50000 \times 1{\cdot}005^{24}$	i.e. \$56 357·99	
After three years you will owe	$\$50000 \times 1{\cdot}005^{36}$	i.e. \$59 834·03	etc.

Similar repeated multiplication on an annual basis can occur when the value of an asset is **depreciated** each year.

Suppose a company purchases a new machine worth \$100 000. One year later the machine is no longer new so its value will have depreciated. One method used for determining the value at the end of each year is to depreciate the value of the asset by a certain percentage of its value at the beginning of that year. Thus if the machine initially worth \$100 000 is depreciated at an annual rate of 12% its value in later years will be as follows:

Value of the machine after 1 year	$= \$100000 \times 0{\cdot}88$	i.e. \$88 000	
Value of the machine after 2 years	$= \$100000 \times 0{\cdot}88^2$	i.e. \$77 000	(nearest \$1000)
Value of the machine after 3 years	$= \$100000 \times 0{\cdot}88^3$	i.e. \$68 000	(nearest \$1000)
	etc.		

You may recall from Unit 3 of *Mathematics Applications* that the above form of depreciation, called the **reducing balance** method (or fixed percentage method or diminishing value method) is not the only method for calculating depreciation.

We also have: the **flat rate**, or straight line, method, in which the depreciation is a fixed amount each year.

and the **unit cost** method, in which the depreciation is related to the units of production the machine has produced.

For example suppose a machine has an initial value of \$500 000.

Consider the following three depreciation schedules:

Flat rate depreciation of \$50 000 per year.

Initial value	\$500 000			
Value after 1 yr	\$450 000	Depreciation in 1st yr	\$50 000	
Value after 2 yrs	\$400 000	Depreciation in 2nd yr	\$50 000	
Value after 3 yrs	\$350 000	Depreciation in 3rd yr	\$50 000	
Value after 4 yrs	\$300 000	Depreciation in 4th yr	\$50 000	etc.

Reducing balance depreciation of 10% per year.

Initial value	\$500000			
Value after 1 yr	\$500000 × 0·9 = \$450000	Depreciation in 1st yr	\$50000	
Value after 2 yrs	\$450000 × 0·9 = \$405000	Depreciation in 2nd yr	\$45000	
Value after 3 yrs	\$405000 × 0·9 = \$364500	Depreciation in 3rd yr	\$40500	
Value after 4 yrs	\$364500 × 0·9 = \$328050	Depreciation in 4th yr	\$36450	etc.

Unit cost depreciation of \$60000 per 100000 units of production.

Initial value	\$500000	
Value after 100000 units of production	\$440000	
Value after 200000 units of production	\$380000	
Value after 300000 units of production	\$320000	etc

Exercise 3D

1. An asset has an initial value of \$220000. Determine its value at the end of each of the first three years using (a) flat rate depreciation of \$25000 per year,
   (b) reducing balance depreciation of 15% per year.

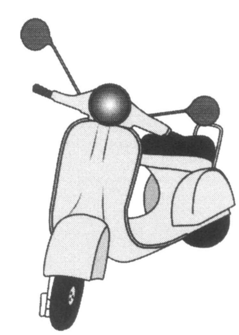
2. An asset has an initial value of \$80000.
   By how much does the asset depreciate in each of its first three years if the depreciation method used is (a) flat rate depreciation of \$10000 per year,
   (b) reducing balance depreciation of 20% per year?

3. A printing machine initially worth \$30000 is expected to be worth just \$2000 after it has been used to produce one million copies.
   Using the unit cost depreciation method what will be the value of the machine after it has produced (a) 250000 copies, (b) 500000 copies, (b) 750000 copies?

4. Some calculators with financial capabilities, and some online financial calculators, can work out depreciation schedules. Investigate.

Loans with regular repayments.

Rather like the investments that involved a regular contribution, people often make regular contributions to reduce the balance of a loan.



For example suppose Jim wants to buy a motor scooter costing \$4000.
Further suppose that Jim borrows the \$4000 from a bank, that he is charged 1% interest per month on the remaining balance and he manages to repay \$100 at the end of every period of one month.

Initially he owes	\$4000		
After 1 month he will owe:	\$4000 × 1·01 − \$100	= \$3940	
After 2 months he will owe:	\$3940 × 1·01 − \$100	= \$3879·40	
After 3 months he will owe:	\$3879·40 × 1·01 − \$100	= \$3818·194	etc,

the value of the loan reducing each month because the amount repaid exceeds the interest charged. Such an arrangement is referred to as a **reducing balance loan**.

Example 8

Johan takes out a loan of \$7000. Interest of 9·4% per annum is added to the loan annually and Johan repays \$800 at the end of each one year period. How much will Johan still owe immediately after he makes the \$800 repayment at the end of year 8?

Initial amount owed:	\$7000	
Amount owed after 1 year:	\$7000 × 1·094 − \$800	= \$6858
Amount owed after 2 years:	\$6858 × 1·094 − \$800	= \$6702·65 (nearest cent)
Amount owed after 3 years:	\$6702·65 × 1·094 − \$800	= \$6532·70 (nearest cent)

We have $T_{n+1} = T_n \times 1{\cdot}094 - \$800$,

and $T_0 = \$7000$

Using the recursion capability of some calculators, or a spreadsheet:

☑ $a_{n+1}=1.094 \cdot a_n-800$
$a_0=7000$
☐ $b_{n+1}=\square$
$b_0=0$
☐ $c_{n+1}=\square$
$c_0=0$

n	a_n
4	6346.8
5	6143.4
6	5920.8
7	5677.4
8	$\boxed{5411.1}$

5411.08529854263

	A	B	C	D
1	Amount borrowed			\$7,000.00
2	Interest rate			9.40
3	Annual repayments			\$800.00
4	Balance at end of year		1	\$6,858.00
5			2	\$6,702.65
6			3	\$6,532.70
7			4	\$6,346.78
8			5	\$6,143.37
9			6	\$5,920.85
10			7	\$5,677.41
11			8	\$5,411.09
12				
13				

The amount still owed at the end of year 8 is \$5411·09.

Note

- Continuing down either of the previous tables allows us to see when the loan will be reduced to zero. In this case the payment at the end of the twentieth year would pay off the loan (and this final payment would only need to be approximately \$200, not \$800).

n	a_n
16	2150.9
17	1553.1
18	899.10
19	183.62
20	$\boxed{-599.1}$

−599.121602634385

- By redefining the recursive rule using a different repayment amount, or by inserting a different repayment amount into the appropriate cell of the spreadsheet, we can investigate how much the annual repayment needs to be in order to pay off the loan in the eight years. Investigate this for yourself and confirm that in this case the required amount is \$1283·58 (with the final repayment being a few cents more).

Once again we could use the financial programs available on some calculators and internet websites that allow the remaining balance of a loan after a number of repayments, or the repayment required to pay off a loan in a particular time etc., to be determined. The display on the right shows the \$1283·58 mentioned in the previous dot point (i.e. the regular repayment needed to pay the loan off in 8 years.) Whilst these programs can be useful make sure that if the course requires you to be able to use spreadsheets or recursive formulae you can do so.

Compound Interest	
N	8
I%	9.4
PV	7000
PMT	−1283.584243
FV	0
P/Y	1
C/Y	1

The display shows that repayments (PMT) of \$1283·584243, i.e. \$1283·58 to the nearest cent, made once per year (P/Y), will see a loan of \$7000 reduced to a future value (FV) of \$0 in 8 years when compound interest is charged at 9·4% per annum, compounded annually, i.e. 1 compounding per year (C/Y). In the display the repayment is shown as a negative - it is money being *taken from us* and put into the account. The \$7000 on the other hand is shown as positive - it is money being *given to us* in the form of a loan.
(Paying \$1283·58 involves rounding the exact payment *down*, hence the final payment will be a few cents more than \$1283·58.)

Exercise 3E

1. Use the display on the right to determine what goes in the blank spaces A to I in the following:

   Repayments of __A__ , i.e. __B__ to the nearest cent, made __C__ times per year, will see a loan of __D__ reduced to a future value of __E__ in __F__ years when compound interest is charged at __G%__ per year, compounded __H__, that is __I__ compoundings per year.

Compound Interest	
N	24
I(%)	9
PV	4000
Pmt	−182.7389691
FV	0
PpY	12
CpY	12

2. At the beginning of one year Jane borrows \$10000. She repays \$2000 at the end of that year and at the end of each year thereafter (except the last) until the loan is paid off. Interest is calculated at 10% per annum, compounded annually. The table below shows the progress of the loan (to the nearest cent).

Year	Amount owed at beginning of year	Amount owed at end of year		
1	\$10000	\$10000 × 1·1 − \$2000	=	\$9000
2	\$9000	\$9000 × 1·1 − \$2000	=	\$7900
3	\$7900	\$7900 × 1·1 − \$2000	=	\$6690
4	\$6690	\$6690 × 1·1 − \$2000	=	\$A
5	\$A	\$A × 1·1 − \$2000	=	\$B
6	\$B	\$B × 1·1 − \$2000	=	\$2284·39
7	\$2284·39	\$2284·39 × 1·1 − \$2000	=	\$C
8	\$C	\$C × 1·1 − \$D	=	\$0

Determine A, B, C and D.

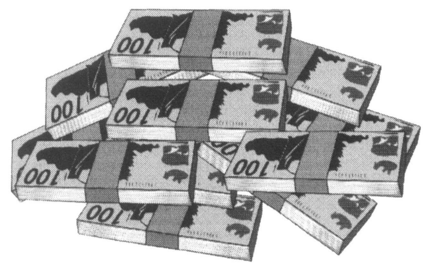
3. Ranii borrows $1 600 at the beginning of one month and repays $300 at the end of that month, and every month thereafter (except the last) until the loan is repaid. Interest is calculated at 1·5% per month.
The table below shows the progress of the loan (to the nearest cent).

Month	Amount owed at beginning of month	Amount owed at end of month		
1	$1600	$1600 × A – $300	=	$1324
2	$1324	$1324 × A – $300	=	$B
3	$B	$B × A – $300	=	$759·52
4	$759·52	$759·52 × A – $300	=	$470·91
5	$470·91	$470·91 × A – $300	=	$177·97
6	$177·97	$177·97 × A – $C	=	$0

Determine the values of A, B and C.

Use recursion techniques for questions 4, 5 and 6.

4. If $*A* is borrowed at a monthly interest rate of 1%, with $*P* being repaid at the end of each month, the amount owing after (n + 1) months is related to the amount owing after n months according to the rule

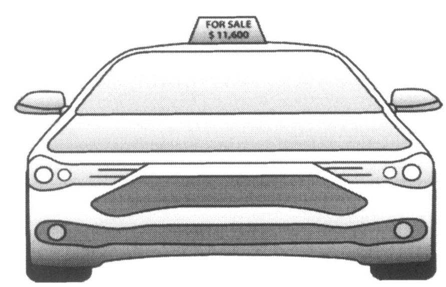


$$T_{n+1} = 1{\cdot}01(T_n) - P, \quad \text{for } n \geq 0,$$

where $T_0 = A$,
and $\$T_n$ is the amount owing after n months.
Find the amount owing at the end of each of the first four months if $50 000 is borrowed at a monthly interest rate of 1% and $800 is repaid at the end of each month.

5. Hisham takes out a loan for $24000 agreeing to repay $350 at the end of each monthly period, after monthly interest is added at 1% of the outstanding balance. How much will he still owe on this loan after 5 years?
Suppose instead that Hisham had made monthly repayments of $375. How much would he still owe after 5 years in this case?

6. Jackson needs to borrow some money from a finance company so that he can purchase a car costing $11600. He is considering three different repayment schemes, all of which involve him in making regular monthly repayments and with interest added at the rate of 15% per year (i.e. 1·25% per month) calculated monthly on the outstanding balance before the repayment is made.



Scheme A: Borrow the full $11600 and make monthly repayments of $250.
Scheme B: Borrow the full $11600 and make monthly repayments of $290.
Scheme C: Pay $1200 that he has saved towards the cost of the car, borrow the rest and make monthly repayments of $290.

How long will each scheme take to pay off the loan?

Use the inbuilt financial programs available on some calculators or a suitable online financial calculator to answer each of questions 7, 8 and 9.

7. To the nearest cent what constant monthly amount needs to be repaid to see a loan of \$10000, with compound interest of 6% per annum compounded monthly, paid off in 2 years?

8. To the nearest cent what constant monthly amount needs to be repaid to see a loan of \$50000, with compound interest of 12% per annum compounded monthly, paid off in 10 years?

9. To the nearest cent what constant monthly amount needs to be repaid to see a loan of \$235000, with compound interest of 9% per annum compounded monthly, paid off in 25 years?

10. The spreadsheet below left is for the situation of borrowing \$4000, paying interest of 1% per month and repaying \$100 per month. The spreadsheet below right shows the same situation but now with \$150 repaid each month.

	A	B	C	D
1	Amount borrowed			**$4,000.00**
2	Interest rate per month			**1.00%**
3	Monthly repayments			**$100.00**
4	Balance at end of month		1	$3,940.00
5			2	$3,879.40
6			3	$3,818.19
7			4	$3,756.38
8			5	$3,693.94
9			6	$3,630.88
10			7	$3,567.19
11			8	$3,502.86
12			9	$3,437.89
13			10	$3,372.27
14			11	$3,305.99
15			12	$3,239.05
16			13	$3,171.44
17			14	$3,103.15
18			15	$3,034.19
19			16	$2,964.53
20			17	$2,894.17
21			18	$2,823.12
22			19	$2,751.35
23			20	$2,678.86
24			21	$2,605.65
25			22	$2,531.70
26			23	$2,457.02
27			24	$2,381.59

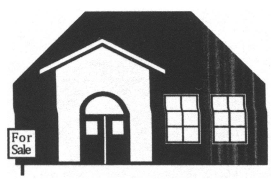
	A	B	C	D
1	Amount borrowed			**$4,000.00**
2	Interest rate per month			**1.00%**
3	Monthly repayments			**$150.00**
4	Balance at end of month		1	$3,890.00
5			2	$3,778.90
6			3	$3,666.69
7			4	$3,553.36
8			5	$3,438.89
9			6	$3,323.28
10			7	$3,206.51
11			8	$3,088.58
12			9	$2,969.46
13			10	$2,849.16
14			11	$2,727.65
15			12	$2,604.92
16			13	$2,480.97
17			14	$2,355.78
18			15	$2,229.34
19			16	$2,101.63
20			17	$1,972.65
21			18	$1,842.38
22			19	$1,710.80
23			20	$1,577.91
24			21	$1,443.69
25			22	$1,308.13
26			23	$1,171.21
27			24	$1,032.92

Create such a spreadsheet yourself, or adapt one you have already created, and find how much needs to repaid per month in order to pay the loan off in exactly two years. Suppose instead the interest rate was 1·2% per month. How much would need to be repaid per month now to repay the loan in exactly two years?

What price house can these people afford?

Let us suppose that in order to purchase a house costing \$480000 John and Maxine pay a deposit of \$30000 from their savings and borrow the remaining \$450000 from a financial institution. The institution charges a fixed 8% per annum compound interest with compounding occurring monthly. John and Maxine arrange to repay the loan by making regular monthly payments for twenty five years.

Using a calculator with the ability to perform financial calculations, or a similar financial capability on a computer or online, we can determine that their monthly repayments would be \$3473·17, to the nearest cent.



Compound Interest	
N	300
I%	8
PV	450000
PMT	-3473.172987
FV	0
P/Y	12
C/Y	12

Before entering into such an arrangement John and Maxine would probably have first worked out what they could afford to pay per month.

Let us suppose that Kym and Hasim have \$35000 saved to put down as a deposit on a house. They feel they can afford to pay \$2000 per month towards the cost of a loan. They plan to get a loan for a term of 25 years making regular monthly payments and are told the interest rate is 7·6% per annum compounded monthly.
What value house can they afford?

As the display on the right indicates, Kym and Hasim can afford a loan of \$268000, to the nearest \$1000.

Therefore, with their deposit of \$35000, they could afford a house costing \$303000.
(= \$268000 + \$35000).

Compound Interest	
N	300
I%	7.6
PV	268273.4425
PMT	-2000
FV	0
P/Y	12
C/Y	12

In the case of Kym and Hasim if they borrow $268 000 they will make 300 repayments each of $1997·96 (see the display on the right).

Hence their total repayments will be

300 × $1997·96 = $599388.

Thus on their loan of $268 000 they have paid

$599 388 − $268 000 = $331388 interest,

... or have they? Read on.

By switching to *"Amortization"*, as shown below right, some calculators show this total interest amount, and other information about the loan. Some calculators, and some online amortization calculators, can display a table showing month by month details about the loan.

Wanting the total interest paid from payment 1 to payment 300 the display on the right shows PM1 as 1 and PM2 as 300 and displays ΣINT, as shown.

If we round the ΣINT amount shown in the display, to the nearest dollar, we again get $331 388.

Wondering where the extra $0·4392 came from?

Well, one calculation used 300 lots of $1997·96 and the other used 300 lots of $1997.961464 ... .

Explore your calculator with regard to amortization and investigate *online* amortization calculators.

Compound Interest	
N	300
I%	7.6
PV	268000
PMT	−1997.961464
FV	0
P/Y	12
C/Y	12

Amortization	
PM1	1
PM2	300
I%	7.6
PV	268000
PMT	−1997.961464
P/Y	12
C/Y	12
BAL	
INT	
PRN	
ΣINT	−331388.4392
ΣPRN	

☞ The slight rounding down of $1997·961464 to $1997·96 means that in theory there would still be a very small amount ($1·3051) remaining on the loan after the 300th payment. In practice, to clear the loan on the 300th payment, the final payment would be $1997·96 + $1·31, i.e. $1999·27.

The total interest amount of $331 388 (= 300 × $1997·96 − $268 000) quoted at the top of the page has not allowed for this change in the final payment. Thus a more accurate calculation of the total interest would be

$331389·31 (= 299 × $1997·96 + 1 × $1999·27 − $268 000),

a total directly obtainable from ΣINT in the amortization calculation with PM1 = 1, PM2 = 300 and monthly repayment of $1997.96.

However, the $331 388 would often be considered a satisfactory approximation.

☞ Under different rounding regimes slight variations are possible. If we round each month to the nearest cent, as some online amortization calculators do, the final payment becomes $1999·06 and the total interest is then

$331389·10 (= 299 × $1997·96 + 1 × $1999·06 − $268 000).

☞ Amortization can also be written amortisation. However, we will use the z spelling here as that is how your calculator is likely to show it.

☞ If you do ever think of taking out a loan to buy a house you should seek expert advice specific to your situation. The work here just introduces the mathematics.

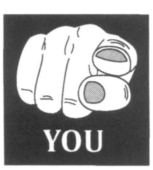
Exercise 3F

What price house can each of the following people afford? In each case assume that the interest rate shown is per annum, compounded monthly.

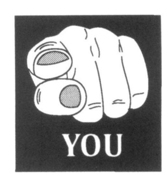
		Amount they can afford to repay per month	Deposit saved.	Duration of loan	Interest rate
1.	Fran and Michael.	\$3800	\$17000	25 years	7·5%
2.	Peta and Peter.	\$2400	\$20000	25 years	8·2%
3.	Rania and Umar.	\$4000	\$35000	20 years	9·1%
4.	Hue and James	\$2000	\$450000	15 years	6·2%
5.	Kirra	\$2650	\$30000	30 years	7·9%

6. Juan and Denise borrow \$450 000 at a fixed interest rate of 8·34% per annum, compounded monthly. They plan to pay off the loan in exactly 25 years by making the same monthly repayments for 25 years.
   How much will each monthly repayment be (nearest cent) and how much interest will they pay in total (nearest \$10)?

7. Chris and Terri borrow \$380 000 at a fixed interest rate of 7·2% per annum, compounded monthly.
   They plan to repay the loan by making the same regular repayments each month for twenty years.
   How much will each repayment be (nearest cent) and how much will they repay in total (i.e. interest plus loan repayment) over the twenty years, to the nearest \$10.
   How much quicker would they have paid off the loan if they had paid \$100 per month more?



In the above cases we have only considered the cost of the house. However other expenses are incurred when purchasing a house, for example stamp duty, insurance, settlement agent fees, purchaser's share of the rates etc. Investigate and write a brief report.



Miscellaneous Exercise Three.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

1. \$8000 is invested for 5 years with an annual compound interest rate of 8%. Find the amount this account is worth at the end of the 5 years if the compounding occurs
   (a) annually, (b) quarterly, (c) monthly.

2. What does it mean if the seasonal index for Autumn is 1·08 (or 108%)?

3. Deseasonalise the following raw data given the seasonal indices are as stated. Give the deseasonalised data to the nearest integer.

Season	Mon	Tue	Wed	Thur	Fri	Sat
Raw data	113	79	84	151	151	160
Seasonal index	0·92	0·70	0·68	1·21	1·29	1·20

4. Find the first five terms of each of the following recursively defined sequences.

(a) $T_1 = 7$, $T_{n+1} = T_n + 12$. (b) $T_1 = 100$, $T_{n+1} = T_n - 15$.

(c) $T_1 = 5000$, $T_{n+1} = 1{\cdot}2 \times T_n$. (d) $T_1 = 2000$, $T_n = T_{n-1} \div 10$.

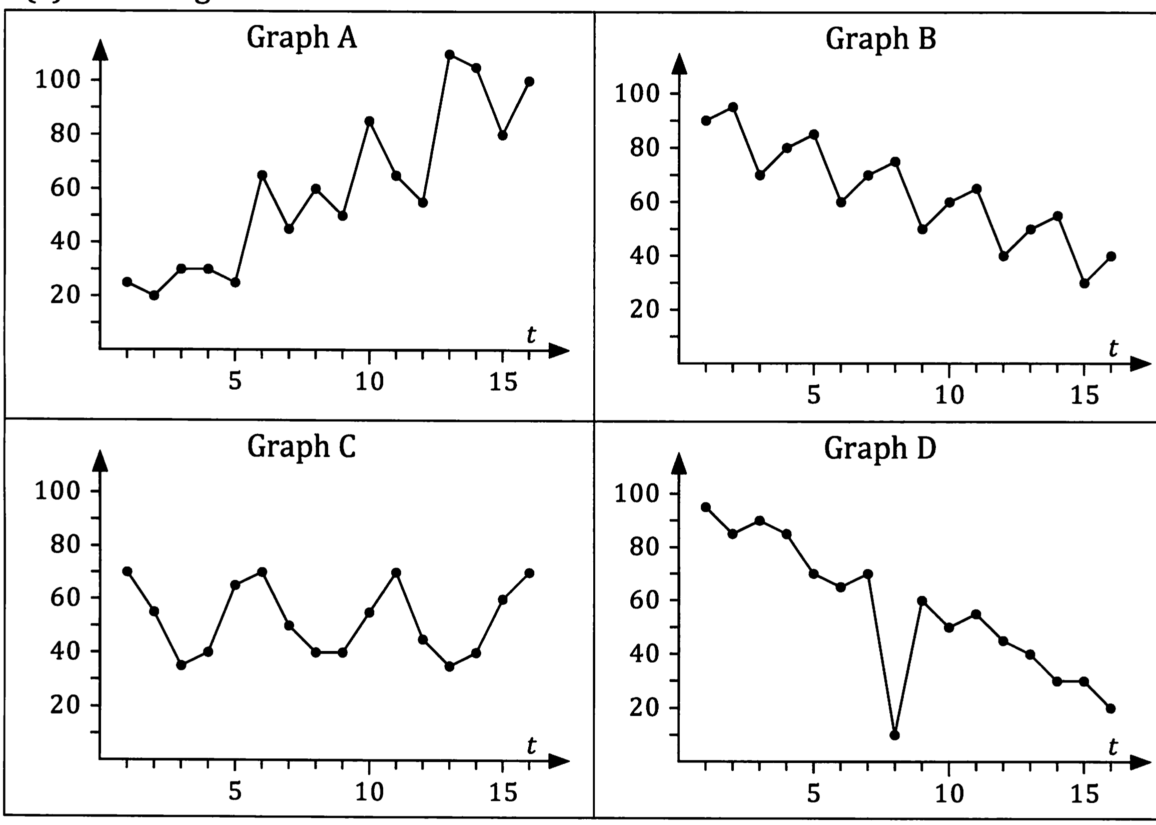
(e) $T_1 = 4$, $T_n = 2T_{n-1} + 3$. (f) $T_1 = 200$, $T_{n+1} = 1{\cdot}5T_n - 4$.

5. Which of the following graphs show

(a) an outlier, (b) seasonal variation,

(c) an increasing trend, (d) a decreasing trend,

(e) no irregular fluctuations?



6. Use recurrence relation techniques for this question.
Johan borrows \$50000 with interest charged at 7·5% per annum on the declining balance, compounded monthly. At the end of the first period of one month, and at the end of every period of one month thereafter, Johan repays \$1000 on the loan. How much will Johan still owe on this loan immediately after the interest has been added and the \$1000 has been credited for the end of

(a) month two, (b) month six, (c) year one, (d) year two?

7. With the aid of a calculator with a built in facility to carry out compound interest calculations, or otherwise, determine each of the following.
   (a) The value at the end of five years of an initial investment of \$12000 invested at 6·4% per annum compound interest, compounded quarterly.
   (b) The number of years required for an initial investment of \$25000 invested at 7·5% per annum compound interest, compounded annually, to be worth \$75000.
   (c) The number of years required to pay off a loan of \$400000 if interest of 8% per annum is compounded monthly and \$3500 is paid off the loan at the end of every month.
   (d) The annual (nominal) compound interest rate required to see an initial investment of \$25000 exceed \$125000 within fifteen years if interest is compounded (i) annually,
   (ii) monthly.

8. In 1971 one Australian dollar, A\$, could purchase approximately 400 Japanese Yen. In 2012 one Australian dollar could purchase approximately 80 Japanese Yen.
The average numbers of Yen, ¥, that could be purchased for each 1 A\$ in each of the years from 1971 to 2012 were approximately as follows:

Year	1971	1972	1973	1974	1975	1976
1 A\$ would buy	400 ¥	372 ¥	367 ¥	408 ¥	405 ¥	379 ¥

Year	1977	1978	1979	1980	1981	1982
1 A\$ would buy	331 ¥	274 ¥	227 ¥	260 ¥	248 ¥	257 ¥

Year	1983	1984	1985	1986	1987	1988
1 A\$ would buy	233 ¥	211 ¥	193 ¥	141 ¥	101 ¥	100 ¥

Year	1989	1990	1991	1992	1993	1994
1 A\$ would buy	107 ¥	112 ¥	107 ¥	100 ¥	83 ¥	73 ¥

Year	1995	1996	1997	1998	1999	2000
1 A\$ would buy	70 ¥	77 ¥	90 ¥	86 ¥	78 ¥	68 ¥

Year	2001	2002	2003	2004	2005	2006
1 A\$ would buy	61 ¥	66 ¥	70 ¥	79 ¥	80 ¥	85 ¥

Year	2007	2008	2009	2010	2011	2012
1 A\$ would buy	93 ¥	99 ¥	75 ¥	81 ¥	82 ¥	81 ¥

(ABARES 2013, Agricultural commodity statistics 2013. CC BY 3.0)

With the assistance of a computer spreadsheet, display the above data graphically, together with the five point moving averages.
Write a few sentences summarising the data.
How does today's exchange rate compare?